

REPORT OF INDIVIDUAL DESIGN PROJECT

INTAKE 40

INTELLIGENT ELECTRIC FENCING FOR ANIMALS

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ABSTRACT

The growing outbreak of human-animal conflict in farm fields has led to extensive damage of crops and huge economic losses to farmers. In Sri Lankan and global perspectives, traditional electric fencing systems have been ineffective in terms of adaptability and efficiency, particularly within areas where several species of animals meet the protected region. This research proposes a smart electric fence system with the ability to automatically identify and detect animal species using computer vision and machine learning. The system uses a YOLOv5n object detection

model trained on a localized dataset featuring images of elephants, cows, and squirrels in an attempt to distinguish between animals in real-time with high accuracy. Based on the identified species, output voltages are dynamically controlled using buck converters with 6V for elephants, 3V for cows, and activating a non-lethal buzzer alarm for squirrels. The response in question minimizes unnecessary damage without compromising deterrence. The entire system is implemented on a Raspberry Pi platform, integrating image processing, object detection, and voltage

control within an embedded solution. It focuses on the feasibility of combining AI-driven animal detection with

control systems embedded in a product to create a cost-effective, scalable, and intelligent solution for minimizing crop damage, reducing manual labor dependency, and facilitating wildlife-agriculture coexistence.

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CHAPTER 1: INTRODUCTION

Wildlife encroachment continues to be a major problem for agriculture, causing severe crop loss and raising labor expenses for manual monitoring. Despite their widespread use, traditional electric fences are ineffective in dynamic farm contexts due to their lack of intelligence and adaptability [1].

Smart solutions are revolutionizing field protection for farmers thanks to developments in computer vision, artificial intelligence, and embedded technologies. Using a Raspberry Pi and the YOLOv5 object detection model, this project suggests an intelligent electric fence system that can recognize and categorize animals like squirrels, cows, and elephants in real time. Buck converters and GPIO-based controls are then used by the system to automatically modify its response [5].

This technique minimizes reliance on manual labor, increases efficiency, and offers a scalable, economical approach to contemporary crop security by combining automated deterrent mechanisms with real-time object detection.

1.1: PROBLEM STATEMENT AND SOLUTIONS

Classic electric fences have a constant voltage, generally ignoring the variability of the threats provided by various animal species. Small animals can easily bypass the fence, and larger animals might not be sufficiently deterred. Inappropriate voltage levels cause crop destruction, loss of livestock, or even possible damage to animals. Manual monitoring of such systems minimizes efficiency and maximizes labor expenses.

This study suggests a smart electric fence system that employs deep learning for real-time animal identification and classification in a bid to counter such challenges. The system adapts in terms of response of the fence at all times based on the species identified, issuing buzzers for small animals and voltage control for large animals. This independent system guarantees animal-friendly operation, increases security, and requires less human intervention.

1.2 : PROJECT OVERVIEW AND SCOPE

By automatically recognizing approaching animals and reacting appropriately, this project seeks to create an intelligent electric fence system that improves farm security. The system uses embedded technology and deep learning to provide adaptive control mechanisms and real-time animal detection.

Three main parts make up the project- a dynamic voltage control unit to safely deter larger animals, an alternative repellent mechanism for smaller animals, and an animal identification system that uses YOLOv5n for precise classification. This intelligent technology guarantees effective protection of crops and livestock, decreases the need for manual monitoring, and enhances wildlife safety. For practical agricultural applications, the system is made to be both affordable and scalable.

1.2.1 : The Animal Detection System

The function of the animal detection system is to utilize computer vision and deep learning technology to detect small and large animals that are approaching the fence. Detection of animals like squirrels, cows, and elephants in real-time with high precision is the main task of the system. The system achieves real-time video input through assistance from the YOLOv5n model and provides a real-time prediction to enact. High detection efficiency, quick processing rate, resistance to environmental fluctuations, simplicity of deployment, and flexibility to handle farm sizes of varying capacities are some of the salient goals of the system.

1.2.2 The Embedded Control and Response System

Real-time animal detection data is analyzed by the embedded control system, which further adjusts the electric fence suitably. It uses a Raspberry Pi to identify the detected animals, run the YOLOv5n deep model, and activate the relevant deterrent mechanism. According to the classification, a non-lethal buzzer for small animals is activated and a buck converter for modifying the fence voltage for large animals is utilized. The system provides automatic decision-making without any human involvement with dynamic control, energy efficiency, and safety.

CHAPTER 2: PRODUCT SURVEY

Many commercial electric fencing systems, including Speedrite and Zareba®, are utilized extensively in agriculture to discourage animals through fixed-voltage shock systems. Although they are fit for general protection, they are not able to differentiate between various species of animals or respond accordingly. They are mostly not so effective for farms dealing with a variety of wildlife problems since they incorporate motion detectors or even manual adjustments. In addition, small animals that are capable of penetrating conventional fences are not effectively addressed using current measures. This calls for a high-tech, artificial intelligence (AI)-driven system with adaptive voltage regulation, deterrent strategies, and real-time animal detection—each of which forms part of the suggested smart electric fence system.

2.1 COMPARISON, RELEVANCE AND KEY DIFFERENCES

Comparison and Relevance to Existing Electric Fence Systems

Although commercially available electric fencing systems, like Zareba® and Speedrite, provide good animal deterrents, they lack intelligent animal recognition and have fixed-voltage outputs. Although these systems provide general protection, they are not suited to distinguish between animal species or vary their response based on that. The suggested smart electric fence system, on the other hand, incorporates embedded control and deep learning for the real-time identification of animals and classification of certain animals, which is a more intelligent and adaptive solution appropriate for dynamic farm environments.

Key Differences of the Custom Smart Electric Fence System

This study provides an intelligent approach that modifies the fence response dynamically according to the type of animal present, unlike traditional systems. The small animals are repelled by a non-lethal repellant, and a voltage controller system is activated by big animals. This customized approach minimizes damage as it improves security, providing improved safety and efficiency. In addition to the static properties of conventional electric fences, the system provides an individualized, context-aware solution through the integration of YOLOv5n based detection and real-time integrated decision-making.

CHAPTER 3: OBJECTIVES AND AIMS

3.1 THE ANIMAL DETECTION SYSTEM

3.1.1 Aim of the Animal Detection and Smart Fencing System

Use deep learning and computer vision techniques to identify and classify different animals approaching the electric fence. Then, depending on the type of animal, execute the corresponding deterrent action.

3.1.2 Objectives of the Disease Detection System

- Recognizing and classifying various species of animals accurately (e.g., elephant, cow, squirrel)
- Low-latency real-time detection
- Implanted devices to automatically control fence voltage and repellents

3.2 THE SMART VOLTAGE CONTROL AND RESPONSE SYSTEM

3.2.1 Aim of the Smart Voltage Control and Response System

Integrate deep learning-based animal identification with integrated control systems to provide automatic, real-time voltage adjustment and animal deterrent responses.

3.2.2 Objectives of the Smart Voltage Control and Response System

- Smooth interface to the animal detection system to allow species-specific response action
- Utilize Arduino-powered relays and buck convertors to regulate output voltage levels based on the animal type.
- For small animals, turn on extra deterrent devices (buzzer) to stop invasions without causing damage.
- Electric fence systems can be operated more efficiently and less often need human intervention
- Implement scalable and reliable operation appropriate for continuous field deployment in agriculture

CHAPTER 4: METHODOLOGY

4.1 : AUTOMATED RESPONSE CONTROL UNIT

4.1.1 : Designing Principle of the Automated Response Control Unit

The Arduino Uno serves as the central unit of this project's automation system, which includes essential elements like relay modules, buck converters, a buzzer, and a 12V power supply. Once the animal classification data from the detection system has been processed, the Arduino drives the relays to activate the corresponding buck converters, which accordingly modify the voltage of the electric fence. For smaller animals, a non-lethal deterrent buzzer is activated. This hardware integration provides for the secure, energy-efficient operation of the electric fence system, dynamic voltage control, and effective deterrent.

4.1 : Flow Chart OF Automated Response Control Unit

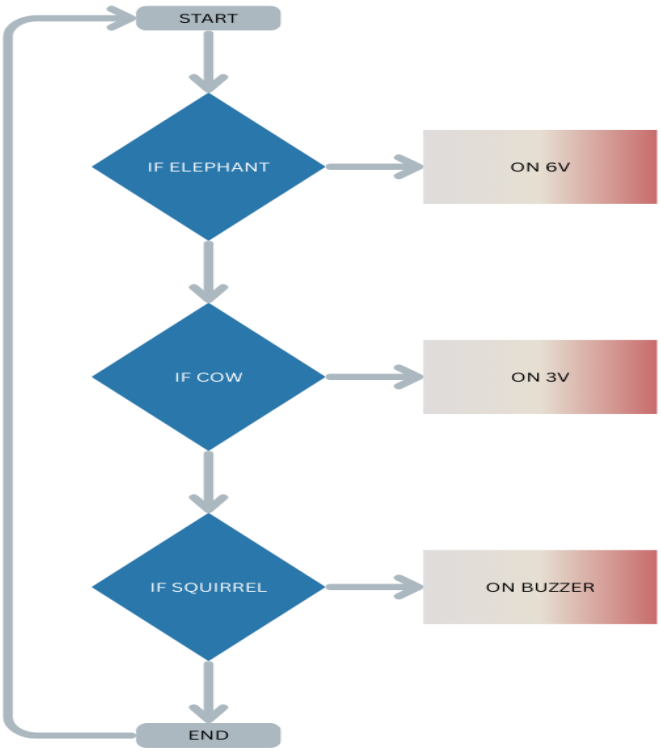


Figure 1: Flow Chart of Automated Response Control Unit

4.2 : Components of the Automated Response Control Unit

- **Arduino Uno**

The ATmega328P microcontroller powers the Arduino Uno, which serves as the master controller of the intelligent electric fence system. It interprets data from the animal detection module and executes commands to activate individual deterrent systems. It is best suited to command real-time voltage control and alarm systems because it is very flexible to be used with different modules and sensors.



Figure 2: Arduino Uno

- **Buck Converters**

The 12V DC power is stepped down to the different voltage levels needed for the electric fence by buck converters. Each of the buck converters, managed by the Arduino and relay module, is accompanied by a voltage level that can be used for repelling a particular animal species. This is a safe and efficient method of adjusting the fence output to the target animal.



Figure 3: Buck Converters

- **Relay Module**

Based on the type of animal detected, the Arduino toggles the relay module, which is an interface of switches to enable the right buck converter. There is a buck converter of a particular voltage rating on every relay. The relay provides the fence system with the right voltage to act effectively as a deterrent when triggered.

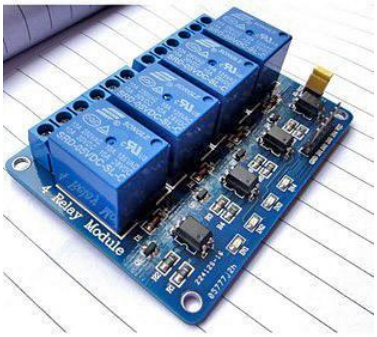


Figure 4: Relay module

- **Buzzer**

The Arduino controls the buzzer directly, and it beeps when it detects small animals, like squirrels. As a non-lethal deterrent, the buzzer beeps rather than shocking you with electricity. This method has the effect of keeping away small animals from the secure area and also improves system safety.



Figure 5: Buzzer

- **12V DC Power Supply**

For buck converters, the relay module, and other electronic devices, the input voltage is supplied by the 12V power supply. For real-time system operation, it provides smooth operation and reliable supply of power.



Figure 6: 12V DC Power Supply

4.3: DATA COLLECTION PART

4.3.1 : Initial Data Collection

A set of 600 images with 200 each of the three classes of animal's squirrels, cows, and elephants were gathered from a location in Kegalle. The images were taken from a wide variety of viewpoints and under various lighting and environmental conditions to make the dataset stronger and more diverse. The deep learning-based detection system can generalize and detect animals approaching the electric fence more effectively in real-world situations owing to this diversity.



Figure 7: Three Animals

4.3.2 Bounding Box Process

Label Studio was utilized to manually draw bounding boxes around the 600 images gathered of squirrels, cows, and elephants. For the model to learn the position and the shapes of the animals well, the bounding box for every one of them was drawn to cover the part of the animal that is visible in the image. To develop a competent deep learning model to detect animals, this step provided proper labeling.

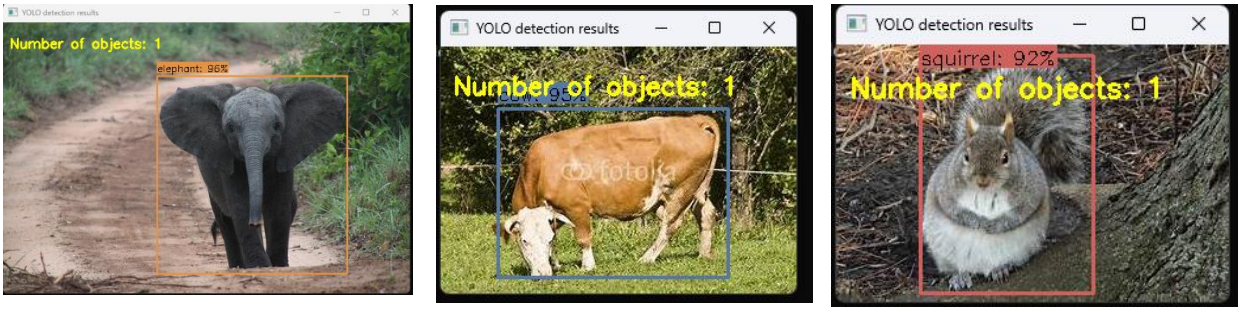


Figure 8: Bounded box images

4.3.3 : Image Preprocessing

All 600 gathered images were preprocessed to have a uniform size, quality, and format to get them ready for training. The photos were resized to a uniform 480x480 pixel size to give a uniform input size appropriate for the YOLOv5n model. Uniformity during model training was ensured because of this resizing

4.4: MODEL PREDICTION UNIT

4.4.1 : Designing Principle of Model Prediction Unit

The project employs an HD webcam and a Raspberry Pi 3B to develop an automatic animal detection system and an electric fence control system. The YOLOv5 model architecture improved by transfer learning is utilized for animal detection, while Python is utilized for system control. A specific dataset separated into training and validation sets is utilized to train the YOLOv5 model for more than 40 epochs. The model performance is tested and optimized, if required, upon training. The Raspberry Pi uses the trained model to offer real-time animal detection and voltage control subsequently.

4.4.2 : Components of the Model Prediction Unit

- **Raspberry Pi 3 Model B**

The core of this system's Model Prediction Unit is Raspberry Pi 3 Model B, a compact minicomputer that is powerful but tiny. It contains microSD card storage and 1GB RAM. In addition to using an HD webcam to capture images, this unit utilizes the trained YOLOv5n model for animal detection in real time and for regulating the voltage of the electric fence



Figure 9: Raspberry Pi 3 Model B

- **Full HD Webcam**

High definition, clear images of animals are captured by a full HD webcam so that the trained YOLOv5n model can accurately recognize them. The technology can specifically differentiate between various animals since the camera is positioned in a manner that it provides the best possible view of the area to be monitored. Accuracy and efficiency are maximized in this arrangement because the smart electric fence system can react accordingly based on the animal detected.



Figure 10: Full HD Webcam

4.5 PRODUCT PARTS DESIGN

Enclosure Box Design

The SolidWorks software was utilized in designing the enclosure box, into which all project components—Raspberry Pi, relay module, buck converters, and power supply—were integrated in a systematic approach. Proper routing, spacing, and ventilation of wires were provided by the design for efficient usage. The 3D model was perfected, and the box was then printed with a 3D printer, yielding a compact and systematic structure suitable for the system configuration.

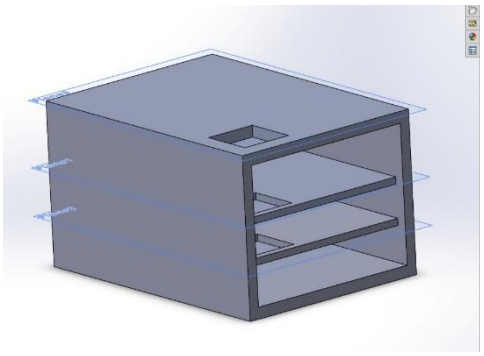


Figure 11: Enclosure Box Design

4.6: COST ANALYSIS

Table 3: Cost Analysis

Item	Amount (Rs.)
Arduino board	1,500.00
Raspberry pi 3B module	22,000.00
USB HD camera	1,200.00
Relay Module	500.00
Power supply	1,200.00
Buck Converters	600.00
Total Amount	**27040.00**

CHAPTER 5: FINAL OUTCOME AND DISCUSSION

5.1 FINAL OUTCOMES

Model outputs



Figure 12: Model Outputs

Product design



Figure 13: Product Design

5.1 DISCUSSION

The major aim of this research is the utilization of computer vision and deep learning algorithms in the detection and classification of various animals. The YOLOv5n model was used for the classification of animals in real-time in real-world applications because of its state-of-the-art object detection and good balance between accuracy and speed.

5.1.1 Model Performance

To find out how well the YOLOv5n model performed in detecting and classifying animals for the smart electric fence system, several important assessment criteria were used.

Confusion Matrix Analysis

Elephants, cows, and squirrels were among the animals the YOLOv5n model for application in the smart electric fence system had extremely high success in detecting and classifying. The confusion matrix, which shows data on the accuracy of the classification provided by the model, is a primary measure of this success. Just one cow was misclassified as background, and all 15 elephants and cows were correctly identified with the matrix. Three of the squirrels were incorrectly labeled as background, and 22 of the 25 occurrences were correctly identified. Consistent with this, the model is working well even when encountering small animals like squirrels but at a bit higher misclassification levels despite being exceedingly excellent on bigger animals like elephants and cows.

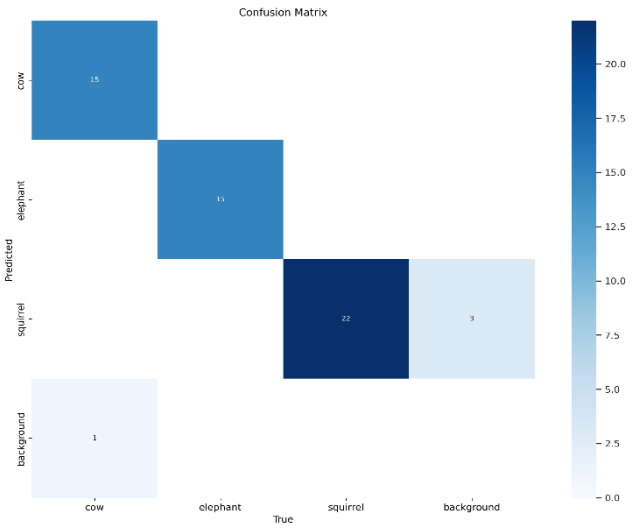


Figure 14: Confusion Matrix

F1-Confidence Curve

The model's F1 score also differs at various confidence levels as can be seen from the F1-confidence graph. The best deployment point is obtained when it reaches a peak F1 score of 0.97 at a confidence level of 0.574. The best threshold maximizes both recall and precision by equating the two since it ensures that the model has tagged most of the animals with a minimal number of false positives. Particularly, although with slight difference in threshold, all classes showed robust F1 scores over a wide range of confidence intervals, proving consistent model performance.

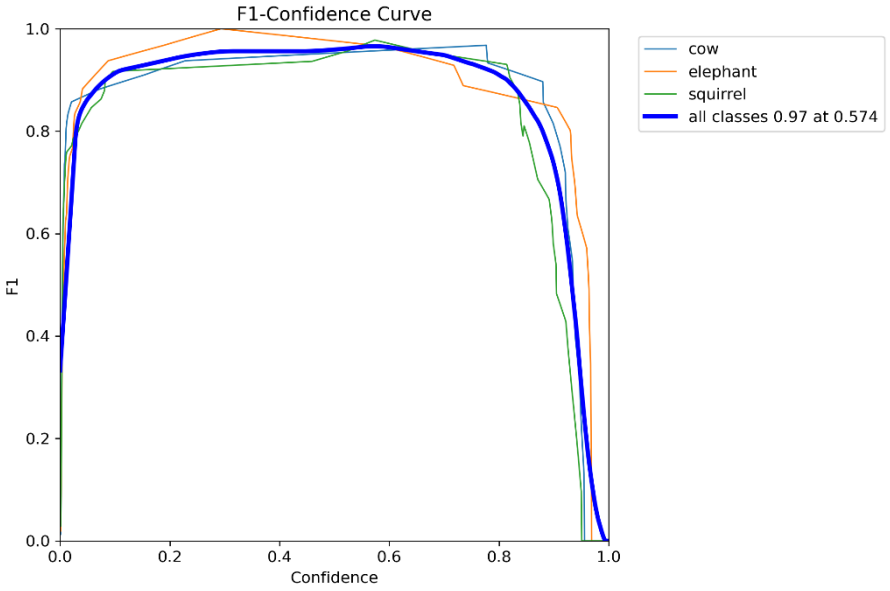


Figure 15: F1 Confidence Curve

Training and Validation Performance

Good training was demonstrated by training loss curves of box_loss, cls_loss, and dfl_loss, all of which showed consistent drops (e.g., box_loss dropped from 1.1 to 0.6). Validation loss followed the same trend, which indicated the model overfit and generalized well.

Precision, recall, and mAP improved consistently across epochs with a maximum at epoch 40. Consistent improvement in metrics indicates that the model converged to a good point and the training was properly optimized.

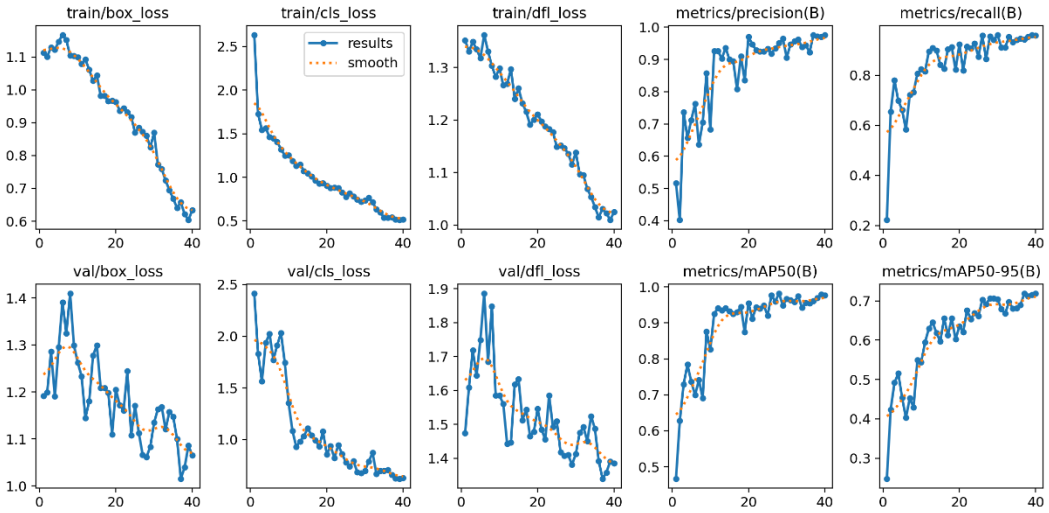


Figure 16: Training and Validation Loss

Visualization and Analysis

The performance of the YOLOv5n model in the smart electric fence system was evaluated by precisely defining the bounding boxes across the detected animals and classifying them accurately into predefined classes like elephant, cow, and squirrel. Further, the capability of the model to detect animals of different sizes was evaluated, demonstrating its strength in detecting animals in a variety of conditions in real-time.



Figure 17: Predetermined groups

5.1.2 Automation System

For real-time species identification, this project's automation system integrates harmoniously with the deep model. In accordance with the classification output of the YOLOv5n model, it regulates the activation of relay modules and buck converters. The system triggers a specific buzzer response or voltage level when detecting a particular species, e.g., an elephant, cow, or squirrel. The successful interaction coordination of the model and hardware units illustrates the system's usability and reliability for intelligent electric fence operation.

5.2 ENCOUNTERED PROBLEMS & IMPLEMENTED SOLUTIONS

1. Delay in Model Predictions due to Limited Processing Power

Problem - Because of the Raspberry Pi 3B's insufficient processing power, using the YOLOv5n model resulted in observable delays during real-time inference.

Solution -

Although the Raspberry Pi 3B was used to finish the project, it was found that a Raspberry Pi 4 or 5 could greatly enhance performance by having improved CPU and RAM functionality. It is suggested that this be updated in the future for more effective function.

2. Limited Dataset Size

Problem - The model's capacity to generalize may be restricted by a small dataset.

Solution - 600 of the Kaggle images, 200 of which were elephants, cows, and squirrels, were utilized in the balanced dataset. This enhanced the model's resilience in various classes. The model could be further improved in performance with the addition of more pictures for each class, though it would offer training examples that are more diversified with more accuracy and stronger generalization.

3. Low FPS Rating Due to Processing Limitations

Problem - The low processing capability of the Raspberry Pi 3B was the major contributor to the model's low frames per second (FPS) when inferring in real-time. This took a toll on the system's responsiveness and real-time detection performance.

Solution - A potential solution to this issue is a transition to a Raspberry Pi 4 or Raspberry Pi 5. These are much more powerful units, which would boost FPS as well as overall system performance, making them more suitable to real-time uses like our project.

5.3 LIMITATIONS OF THIS PRODUCT

1) The model is presently trained only to identify and categorize these three animals because of the limited class-specific dataset, which was 600 images (200 images for each elephant, cow, and squirrel) gathered from Kaggle. Therefore, it is unable to identify other animals or items present in a real environment.

2) The limited processing capacity of Raspberry Pi 3B limits the real-time aspect of the system. Real-time or fast detection is impaired by the low frame-per-second (FPS) output rate being produced using prediction. Real-time efficiency could be enhanced by upgrading to a stronger board like Raspberry Pi 4 or 5.

5.5: KNOWLEDGE & SKILLS GAINED

I have gained a variety of technical abilities and really expanded my knowledge over this endeavor.

1) Object Detection and Model
   I gained a good grasp on how to identify objects in real-time with the YOLOv5n model. Preparing datasets, annotating with Label Studio, training, evaluation, and performance tuning based on critical metrics such as mAP, precision, and recall were part of this.

2) Hardware Integration and System Automation
   My skill with the Raspberry Pi 3B has improved, which was needed to automate the detection process and deploy the trained model. To ensure that animal detection and voltage regulation of the electric fence system were accurate, it needed to include the webcam and manage real-time image capture.

3) 3D Design
   I improved my SolidWorks skills.

4) Electrical and Power System Management
   I gained real-world experience in choosing and controlling power supplies and connections to get the Raspberry Pi and included components to operate smoothly. This includes regulating the power levels of the smart electric fencing system in relation to animal

detection.

5) The ability to program was enhanced.

CHAPTER 6: CONCLUSIONS AND FUTURE WORKS

5.1 : CONCLUSION

This study shows how embedded systems and deep learning can be utilized efficiently to develop an intelligent electric fencing system through real-time identification of animals. Real-time voltage fluctuation as a guided deterrence and effective animal classification were attained using a YOLOv5n model training and deployment on a dataset of 600 images of elephants, cows, and squirrels using a Raspberry Pi 3B. A custom-designed 3D-printed enclosure safely contained all the electronic devices, and careful power management guaranteed steady operation. The model produced consistent classification despite the limited dataset size, and experimentation confirmed its practicality. This study demonstrates the potential of AI-enhanced autonomous animal management systems and provides a solid foundation for field implementation in intelligent agriculture and animal welfare.

5.2 : FUTURE WORKS

1. Developing a Deep Learning Model with Greater Accuracy-By collecting more data sets.

2. Increasing Processing Capacity-by using Raspberry pi 4 or 5.

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ANNEXTURES

Arduino Code for Automation part

```
unsigned long lastSignalTime = 0;
const unsigned long signalTimeout = 2000; // 2 seconds timeout

void setup() {
  Serial.begin(9600);
  pinMode(2, OUTPUT); // V1
```

```
  pinMode(3, OUTPUT); // V2
  pinMode(4, OUTPUT); // V3
}

void loop() {
  // Check for new serial input
  if (Serial.available()) {
    String input = Serial.readStringUntil('\n');
    input.trim(); // Remove extra whitespace

    // Reset all outputs first
    digitalWrite(2, LOW);
    digitalWrite(3, LOW);
    digitalWrite(4, LOW);

    // Set the selected output HIGH
    if (input == "V1") {
      digitalWrite(2, HIGH);
    } else if (input == "V2") {
      digitalWrite(3, HIGH);
    } else if (input == "V3") {
      digitalWrite(4, HIGH);
    }

    // Update last signal time
    lastSignalTime = millis();
  }

  // If no signal received for a while, turn off all outputs
  if (millis() - lastSignalTime > signalTimeout) {
    digitalWrite(2, LOW);
    digitalWrite(3, LOW);
    digitalWrite(4, LOW);
  }
}
```